Figure 17.7 To understand how the parameter α controls cooling in simulated annealing, let's take a look at the *exponential decay* function: `f(i)=`T_i`=α*f(i-1),∀ i>0`. In this chart, we set `f(0)=`T_0`=100`, and show how the function's shape changes with the rate α, which controls how fast the function goes to zero. Why it's called *exponential* decay, you ask? Because $T_1 = \alpha T_0$, $T_2 = \alpha T_1 = \alpha^2 T_0$, **and in general** $T_n = \alpha^n T_0$**: if α is between `0` and `1`,** α^n **and in turn** T_n **get progressively (exponentially) smaller.**

The last thing we need to discuss is how to update temperature. Similarly to the acceptance probability, there are several possible viable options for this function – there are no restrictions on it.

Typically, however, geometric (aka exponential) decay is used, with the temperature value updated not every iteration, but after some interval (for instance, every 1000 iterations or so).

The mathematical formula for this function is:

$$T_i = \alpha T_{i-1}, \; 0 < \alpha < 1$$

The temperature at iteration `i` is a fraction of the temperature at iteration `i-1`: α, which must be between `0` and `1`, controls how cooler temperature gets between two iterations.

Now, you might have to fiddle a bit with the value of α and the interval between temperature updates to tune them and